

Patent Boxes, Tax Credits, or Both?

Michael Devereux^{*} Ben Lockwood[†] Erez Yerushalmi[‡]

1 Introduction

This Online Appendix documents the numerical implementation of the analytical framework developed in the paper. It provides the complete model equations together with the GAMS codes for both the Base Model and the Shifting Model used in the numerical analysis. The material is intended to facilitate replication and future extensions of the model.

2 The setup of the simulated models

This section documents the model setup common to both the Main Model and the Shifting Model (in Section 6) of the paper.

2.1 Firms in the real side of the economy

The real side of the economy is the same in both models. Intermediate firms face inverse demand

$$p_i = A_i q_i^{-1/\varepsilon}, \quad A_i = a_i \left(\frac{\varepsilon - 1}{\varepsilon} \right), \quad \varepsilon > 1, \quad (1)$$

^{*}Michael.Devereux@sbs.ox.ac.uk, Oxford University Centre for Business Taxation, Saïd Business School.

[†]B.Lockwood@warwick.ac.uk, University of Warwick.

[‡]Erez.Yerushalmi@bcu.ac.uk, Birmingham City University Business School.

and maximise pre-tax profits

$$\pi_i^I = p_i q_i - \phi_i q_i. \quad (2)$$

Profit maximisation yields the constant-markup pricing rule

$$p_i = \frac{\varepsilon}{\varepsilon - 1} \phi_i. \quad (3)$$

Substituting the equilibrium quantity into profits gives

$$\pi_i^I = Z a_i^\varepsilon \phi_i^{1-\varepsilon}, \quad Z \equiv \varepsilon^{-2\varepsilon} (\varepsilon - 1)^{2\varepsilon-1}. \quad (4)$$

Let the pre-innovation marginal cost be normalised to one, and let the post-innovation marginal cost be $\phi < 1$. Under symmetry, the pre- and post-innovation intermediate profits are

$$\underline{\pi}^I = Z a^\varepsilon, \quad \bar{\pi}^I = Z a^\varepsilon \phi^{1-\varepsilon}. \quad (5)$$

The innovation gain in intermediate profits is therefore

$$\Delta^I = \bar{\pi}^I - \underline{\pi}^I = Z a^\varepsilon (\phi^{1-\varepsilon} - 1) > 0. \quad (6)$$

Final–rood profits are proportional to intermediate profits:

$$\pi_i^F = m \pi_i^I, \quad m = \frac{\varepsilon}{\varepsilon - 1}. \quad (7)$$

Therefore, the gain in final–rood profits and the total gain from innovation are

$$\Delta^F = m \Delta^I, \quad \Delta = \Delta^I + \Delta^F = (1 + m) \Delta^I. \quad (8)$$

The arm’s-length royalty is pinned down by the Nash bargaining solution:

$$P = \beta \Delta^I, \quad \beta \in (0, 1]. \quad (9)$$

The innovation technology used in the simulations is

$$\sigma = 1 - \exp(-r), \quad r = \alpha \ln l + (1 - \alpha) \ln e, \quad 0 < \alpha < 1. \quad (10)$$

In the Main Model online appendix below we impose symmetry, no spillovers, and work at the individual-firm level, so $n = 1$ and $s = 0$. The same functional forms are used later in the Shifting Model implementation.

2.2 Government problem

The government chooses the policy instruments (τ, c, b) to maximise welfare:

$$\max_{\tau, c, b} W(\tau, c, b), \quad (11)$$

subject to

$$0 \leq \tau \leq \bar{\tau}, \quad 0 \leq c \leq 1 - \tau, \quad 0 \leq b \leq \tau. \quad (12)$$

The upper-bound restrictions can be written as

$$g_1(\tau, c, b) = \tau - \bar{\tau} \leq 0, \quad (13)$$

$$g_2(\tau, c, b) = c + \tau - 1 \leq 0, \quad (14)$$

and

$$g_3(\tau, c, b) = b - \tau \leq 0. \quad (15)$$

The lower bounds $\tau \geq 0$, $c \geq 0$, and $b \geq 0$ are imposed directly as variable bounds in the MCP implementation.

2.3 Lagrangian and MCP representation

Let $\lambda_1, \lambda_2, \lambda_3 \geq 0$ denote the multipliers associated with the three upper-bound constraints. The Lagrangian is

$$\mathcal{L} = W - \lambda_1(\tau - \bar{\tau}) - \lambda_2(c + \tau - 1) - \lambda_3(b - \tau). \quad (16)$$

The stationarity conditions are

$$\frac{\partial \mathcal{L}}{\partial \tau} = \frac{\partial W}{\partial \tau} - \lambda_1 - \lambda_2 + \lambda_3 = 0, \quad (17)$$

$$\frac{\partial \mathcal{L}}{\partial c} = \frac{\partial W}{\partial c} - \lambda_2 = 0, \quad (18)$$

$$\frac{\partial \mathcal{L}}{\partial b} = \frac{\partial W}{\partial b} - \lambda_3 = 0. \quad (19)$$

The complementary slackness conditions are

$$0 \leq \lambda_1 \perp \bar{\tau} - \tau \geq 0, \quad (20)$$

$$0 \leq \lambda_2 \perp 1 - \tau - c \geq 0, \quad (21)$$

and

$$0 \leq \lambda_3 \perp \tau - b \geq 0. \quad (22)$$

Together with the lower bounds on the policy instruments, the stationarity and complementary slackness conditions define the MCP representation:

$$0 \leq \tau \perp -\frac{\partial W}{\partial \tau} + \lambda_1 + \lambda_2 - \lambda_3 \geq 0, \quad (23)$$

$$0 \leq c \perp -\frac{\partial W}{\partial c} + \lambda_2 \geq 0, \quad (24)$$

$$0 \leq b \perp -\frac{\partial W}{\partial b} + \lambda_3 \geq 0, \quad (25)$$

$$0 \leq \lambda_1 \perp \bar{\tau} - \tau \geq 0, \quad 0 \leq \lambda_2 \perp 1 - \tau - c \geq 0, \quad 0 \leq \lambda_3 \perp \tau - b \geq 0. \quad (26)$$

2.4 Model Regimes

Table 1 classifies the optimal policy outcomes into nine regimes, determined by whether the corporate tax rate τ is at an interior optimum or constrained at its upper bound $\bar{\tau}$. In *Regime 1* (τ interior), the government freely optimises over all three instruments and the standard Karush–Kuhn–Tucker first-order conditions hold: the patent box rate b and tax credit c are set

Table 1: Optimal policy regimes: KKT signatures

	τ	b	c	$\frac{dW}{d\tau} _{\text{eff}}$	$\frac{\partial W}{\partial b}$	$\frac{\partial W}{\partial c}$	λ_1	λ_2	λ_3
<i>Regime 1: τ interior ($\lambda_1 = 0$)</i>									
1.1 No instruments	int.	$= 0$	$= 0$	$= 0$	≤ 0	≤ 0	0	0	0
1.2 PB only	int.	$(0, \bar{\tau})$	$= 0$	$= 0$	$= 0$	≤ 0	0	0	0
1.3 PB exhausted	int.	$= \tau$	$= 0$	$= 0^\dagger$	≥ 0	≤ 0	0	0	≥ 0
<i>Regime 2: τ constrained ($\tau = \bar{\tau}$, $\lambda_1 > 0$)</i>									
2.1 No instruments	$\bar{\tau}$	$= 0$	$= 0$	> 0	≤ 0	≤ 0	> 0	0	0
2.2 Credit only	$\bar{\tau}$	$= 0$	> 0	> 0	≤ 0	$= 0$	> 0	$0^\ddagger$	0
2.3 PB interior, $c = 0$	$\bar{\tau}$	$(0, \bar{\tau})$	$= 0$	> 0	$= 0$	≤ 0	> 0	0	0
2.4 PB interior, $c > 0$	$\bar{\tau}$	$(0, \bar{\tau})$	> 0	> 0	$= 0$	$= 0$	> 0	$0^\ddagger$	0
2.5 PB exhausted, $c = 0$	$\bar{\tau}$	$= \bar{\tau}$	$= 0$	> 0	≥ 0	≤ 0	> 0	0	≥ 0
2.6 PB exhausted, $c > 0$	$\bar{\tau}$	$= \bar{\tau}$	> 0	> 0	≥ 0	$= 0$	> 0	$0^\ddagger$	≥ 0

Notes: $\lambda_1, \lambda_2, \lambda_3$ are Lagrange multipliers on the upper-bound constraints $\tau \leq \bar{\tau}$, $c \leq 1 - \tau$, and $b \leq \tau$ respectively; a positive multiplier signals a binding constraint. Lower bounds ($\tau \geq 0$, $b \geq 0$, $c \geq 0$) are imposed as variable bounds in the MCP and carry no separate multiplier. † In Regime 1.3, the constraint $b = \tau$ binds so the relevant optimality condition is the total derivative $dW/d\tau|_{b=\tau} = \partial W/\partial \tau + \partial W/\partial b = 0$; the partial derivatives are individually nonzero and of opposite sign. ‡ In Regime 2.2, 2.4 and 2.6, $\lambda_2 = 0$ throughout all simulations because of A3 and Prop 2.

where their marginal welfare contributions equal zero or, at corner solutions, are non-positive. Three subcases arise depending on whether neither instrument is used (1.1), the patent box alone is active at an interior value (1.2), or the patent box is exhausted at its upper bound (1.3). In *Regime 2*, the binding ceiling $\tau = \bar{\tau}$ prevents the government from freely adjusting the tax rate, and the optimisation reduces to the choice of (b, c) only; six subregimes arise depending on which instruments are active and whether they are interior or at their bounds. The Lagrange multipliers λ_1, λ_2 , and λ_3 – on the constraints $\tau \leq \bar{\tau}$, $c \leq 1 - \tau$, and $b \leq \tau$ respectively – provide the regime signatures reported in the table: a positive multiplier signals a binding constraint whose shadow value measures the welfare cost of the restriction.

Both models share the same analytically predicted regimes when τ is interior (Regime 1). Regime 2 arises when the upper bound $\tau = \bar{\tau}$ binds; it lies outside the hypothesis of Propositions 3 and 6 and is characterised by its subregime structure over (b, c) .

Regime 1 — τ interior ($\lambda_1 = 0$, $0 < \tau < \bar{\tau}$)

Regime 1.1 — No instruments. $b = 0, c = 0$. Both instruments inactive. KKT signature:

$$\frac{dW}{d\tau} = 0, \quad \frac{\partial W}{\partial b} \leq 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 = \lambda_2 = \lambda_3 = 0.$$

The inequalities reflect lower-bound corners for b and c , handled as variable bounds in the MCP. Structurally ruled out for $\delta > 1$ by the surplus gap inequality $\Delta > \Delta^I$.

Regime 1.2 — Patent box only. $0 < b < \tau, c = 0$. Patent box active at an interior value; tax credit inactive. KKT signature:

$$\frac{dW}{d\tau} = 0, \quad \frac{\partial W}{\partial b} = 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 = \lambda_2 = \lambda_3 = 0.$$

Regime 1.3 — Patent box exhausted. $b = \tau, c = 0$. Patent box at its upper bound; tax credit inactive. Since b is pinned to τ , the relevant optimality condition is the total derivative $dW/d\tau|_{b=\tau} = \partial W/\partial\tau + \partial W/\partial b = 0$. KKT signature:

$$\frac{\partial W}{\partial\tau} + \frac{\partial W}{\partial b} = 0, \quad \lambda_3 \geq 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 = \lambda_2 = 0.$$

Regime 2 — τ constrained ($\tau = \bar{\tau}, \lambda_1 > 0$)

The stationarity condition for τ is

$$\frac{\partial W}{\partial\tau} = \lambda_1 + \lambda_2 - \lambda_3,$$

so $\partial W/\partial\tau$ need not equal zero. The government optimises over (b, c) only; six subregimes arise depending on whether b and c are at interior values or at their bounds.

Regime 2.1 — No instruments. $b = 0, c = 0$.

$$\frac{\partial W}{\partial b} \leq 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 > 0, \quad \lambda_2 = \lambda_3 = 0.$$

Structurally ruled out for $\delta > 1$; may arise at the knife-edge $\delta = 1$.

Regime 2.2 — Credit only. $b = 0, c > 0$.

$$\frac{\partial W}{\partial b} \leq 0, \quad \frac{\partial W}{\partial c} = 0, \quad \lambda_1 > 0, \quad \lambda_2 \geq 0, \quad \lambda_3 = 0.$$

Structurally ruled out for $\delta > 1$ by the surplus gap inequality.

Regime 2.3 — PB interior, credit inactive. $0 < b < \bar{\tau}, c = 0$.

$$\frac{\partial W}{\partial b} = 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 > 0, \quad \lambda_2 = \lambda_3 = 0.$$

Regime 2.4 — PB interior, credit active. $0 < b < \bar{\tau}, c > 0$.

$$\frac{\partial W}{\partial b} = 0, \quad \frac{\partial W}{\partial c} = 0, \quad \lambda_1 > 0, \quad \lambda_2 \geq 0, \quad \lambda_3 = 0.$$

Regime 2.5 — PB exhausted, credit inactive. $b = \bar{\tau}, c = 0$.

$$\frac{\partial W}{\partial b} \geq 0, \quad \frac{\partial W}{\partial c} \leq 0, \quad \lambda_1 > 0, \quad \lambda_2 = 0, \quad \lambda_3 \geq 0.$$

The generic subregime for δ sufficiently large.

Regime 2.6 — PB exhausted, credit active. $b = \bar{\tau}, c > 0$.

$$\frac{\partial W}{\partial b} \geq 0, \quad \frac{\partial W}{\partial c} = 0, \quad \lambda_1 > 0, \quad \lambda_2 \geq 0, \quad \lambda_3 \geq 0.$$

Arises when the Pigouvian motive dominates; characterises the knife-edge $\delta = 1$ and low- δ regions of Regime 2.

3 Base Model

3.1 Base Model closed-form input demands

In the Base Model, the R&D firm chooses the observable input l and the unobservable input e . With symmetry and no spillovers, the first-order con-

ditions are

$$\frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial l} \beta \Delta^I = \frac{1 - \tau - c}{1 - \tau + b}, \quad \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial e} \beta \Delta^I = \frac{1}{1 - \tau + b}. \quad (27)$$

It is useful to define the after-tax input prices

$$p_l \equiv \frac{1 - \tau - c}{1 - \tau + b}, \quad p_e \equiv \frac{1}{1 - \tau + b}, \quad (28)$$

and useful to define

$$N \equiv 1 - \tau - c, \quad D \equiv 1 - \tau + b. \quad (29)$$

The functional form of the innovation technology is

$$\sigma = 1 - \exp(-r), \quad r = \alpha \ln l + (1 - \alpha) \ln e, \quad 0 < \alpha < 1. \quad (30)$$

Equivalently, the first-order conditions can be written as

$$\exp(-r) \frac{\alpha}{l} = \frac{p_l}{\beta \Delta^I}, \quad \exp(-r) \frac{1 - \alpha}{e} = \frac{p_e}{\beta \Delta^I}. \quad (31)$$

Solving these two equations gives the closed-form input demands

$$l = \left(\frac{\alpha \beta \Delta^I}{p_l} \right)^{\frac{2-\alpha}{2}} \left(\frac{p_e}{(1-\alpha) \beta \Delta^I} \right)^{\frac{1-\alpha}{2}}, \quad (32)$$

$$e = \left(\frac{p_l}{\alpha \beta \Delta^I} \right)^{\frac{\alpha}{2}} \left(\frac{(1-\alpha) \beta \Delta^I}{p_e} \right)^{\frac{1+\alpha}{2}}. \quad (33)$$

Substituting the definitions of p_l and p_e , these can be written directly in terms of (τ, c, b) :

$$l = (1 - \tau + b)^{1/2} (1 - \tau - c)^{(\alpha-2)/2} (\beta \Delta^I)^{1/2} \alpha^{(2-\alpha)/2} (1 - \alpha)^{(\alpha-1)/2}, \quad (34)$$

$$e = (1 - \tau + b)^{1/2} (1 - \tau - c)^{\alpha/2} (\beta \Delta^I)^{1/2} \alpha^{-\alpha/2} (1 - \alpha)^{(1+\alpha)/2}. \quad (35)$$

The corresponding composite innovation input is

$$r = \alpha \ln l + (1 - \alpha) \ln e, \quad (36)$$

and hence

$$\sigma = 1 - (1 - \tau + b)^{-1/2} (1 - \tau - c)^{\alpha/2} (\beta \Delta^I)^{-1/2} \alpha^{-\alpha/2} (1 - \alpha)^{(\alpha-1)/2}. \quad (37)$$

For the closed-form expressions to be well defined, we require

$$1 - \tau + b > 0, \quad 1 - \tau - c > 0. \quad (38)$$

The second condition is slightly stronger than the weak feasibility constraint $c \leq 1 - \tau$, and is imposed numerically to avoid singular input demands.

3.2 Base Model welfare function

The government objective is pre-tax real surplus plus $(\delta - 1)$ times tax revenue. With symmetry, no spillovers, and $n = 1$, welfare is

$$W = \underline{\pi} + \sigma \Delta - l - e + (\delta - 1) [\tau (\underline{\pi} + \sigma \Delta - l) - b \sigma \beta \Delta^I - cl]. \quad (39)$$

Equivalently, collecting terms gives

$$W = \underline{\pi} [1 + \tau(\delta - 1)] + [(1 + (\delta - 1)\tau) (\Delta^I + \Delta^F) - (\delta - 1)b\beta\Delta^I] \sigma - [1 + (\delta - 1)(\tau + c)] l - e. \quad (40)$$

This is the reduced-form welfare function used in the Base Model simulation.

3.3 Derivative blocks for the Base Model

The derivative structure mirrors the paper appendix. The endogenous objects l, e, r, σ depend on the policy instruments through the two after-tax

input prices p_l, p_e . The price derivatives are

$$\frac{\partial p_l}{\partial c} = -\frac{1}{1 - \tau + b}, \quad \frac{\partial p_l}{\partial b} = -\frac{1 - \tau - c}{(1 - \tau + b)^2}, \quad \frac{\partial p_l}{\partial \tau} = -\frac{b + c}{(1 - \tau + b)^2}, \quad (41)$$

$$\frac{\partial p_e}{\partial c} = 0, \quad \frac{\partial p_e}{\partial \tau} = \frac{1}{(1 - \tau + b)^2}, \quad \frac{\partial p_e}{\partial b} = -\frac{1}{(1 - \tau + b)^2}. \quad (42)$$

The minimal derivative set required for the reduced-form welfare gradients is

$$\frac{\partial \sigma}{\partial r} = 1 - \sigma, \quad \frac{\partial r}{\partial p_l} = -\frac{\alpha}{2p_l}, \quad \frac{\partial r}{\partial p_e} = -\frac{1 - \alpha}{2p_e}, \quad (43)$$

$$\frac{\partial l}{\partial p_l} = -\frac{2 - \alpha}{2} \frac{l}{p_l}, \quad \frac{\partial l}{\partial p_e} = \frac{1 - \alpha}{2} \frac{l}{p_e}, \quad (44)$$

$$\frac{\partial e}{\partial p_l} = \frac{\alpha}{2} \frac{e}{p_l}, \quad \frac{\partial e}{\partial p_e} = -\frac{1 + \alpha}{2} \frac{e}{p_e}. \quad (45)$$

Define A_l, A_e as the tax-revenue response blocks and B_l, B_e as the non-tax welfare response blocks. With symmetry, $n = 1$, and no spillovers, these are

$$A_l = (\tau \Delta - b\beta \Delta^I) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_l} - (\tau + c) \frac{\partial l}{\partial p_l}, \quad (46)$$

$$A_e = (\tau \Delta - b\beta \Delta^I) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_e} - (\tau + c) \frac{\partial l}{\partial p_e}, \quad (47)$$

$$B_l = \Delta \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_l} - \left(\frac{\partial l}{\partial p_l} + \frac{\partial e}{\partial p_l} \right), \quad (48)$$

$$B_e = \Delta \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_e} - \left(\frac{\partial l}{\partial p_e} + \frac{\partial e}{\partial p_e} \right). \quad (49)$$

Using the closed-form derivative expressions, these blocks can also be written as

$$A_l = \frac{1}{2p_l} [-\alpha(\tau \Delta - b\beta \Delta^I)(1 - \sigma) + (2 - \alpha)(\tau + c)l], \quad (50)$$

$$A_e = -\frac{1 - \alpha}{2p_e} [(\tau \Delta - b\beta \Delta^I)(1 - \sigma) + (\tau + c)l], \quad (51)$$

$$B_l = \frac{1}{2p_l} [-\alpha \Delta (1 - \sigma) + (2 - \alpha)l - \alpha e], \quad (52)$$

$$B_e = \frac{1}{2p_e} [-(1-\alpha)\Delta(1-\sigma) - (1-\alpha)l + (1+\alpha)e]. \quad (53)$$

3.4 Base Model welfare derivatives with respect to τ, c, b

Using the price derivatives and the response blocks above, the derivative of welfare with respect to τ is

$$\begin{aligned} \frac{\partial W}{\partial \tau} = & (\delta - 1) \left(\pi + \sigma\Delta - l - A_l \frac{b+c}{(1-\tau+b)^2} + A_e \frac{1}{(1-\tau+b)^2} \right) \\ & - B_l \frac{b+c}{(1-\tau+b)^2} + B_e \frac{1}{(1-\tau+b)^2}. \end{aligned} \quad (54)$$

The derivative with respect to the tax credit is

$$\frac{\partial W}{\partial c} = (\delta - 1) \left(-l - A_l \frac{1}{1-\tau+b} \right) - B_l \frac{1}{1-\tau+b}. \quad (55)$$

The derivative with respect to the patent box is

$$\begin{aligned} \frac{\partial W}{\partial b} = & (\delta - 1) \left(-\sigma\beta\Delta^I - A_l \frac{1-\tau-c}{(1-\tau+b)^2} - A_e \frac{1}{(1-\tau+b)^2} \right) \\ & - B_l \frac{1-\tau-c}{(1-\tau+b)^2} - B_e \frac{1}{(1-\tau+b)^2}. \end{aligned} \quad (56)$$

These expressions are the Base Model counterpart to the Shifting Model derivative system. The difference between the two models lies in the effective input prices, the tax-revenue block, and the welfare treatment of income and cost shifting.

3.5 Variants of Base model - no constraint on b and c

In this variant, we remove constraint on b and c . We maximised welfare $W(\tau, b, c)$, subject to $\tau \leq \bar{\tau}$.

In the original formulation, b and c were constrained by $b \geq 0$, $c \geq 0$, $b \leq \tau$, and $c \leq 1 - \tau$. But when all four of these constraints on b and c are removed, the welfare function has no finite maximum. The reason lies in the structure of the effective input prices that govern firm behaviour:

$$p_l = \frac{N}{D} = \frac{1 - \tau - c}{1 - \tau + b}, \quad p_e = \frac{1}{D} = \frac{1}{1 - \tau + b}. \quad (57)$$

Let's say the model looks for $b+k$ and $c+k$ for any scalar k , $p_l = N/D$ can be exactly unchanged, while $p_e = 1/D$ falls as D grows. Welfare responds to this shift, and crucially it does not converge. W increases (or decreases) without bound as $k \rightarrow \infty$. The two instruments are not independently identified: they share a common direction along which welfare is monotone rather than concave.

GAMS shows this in the output. The solver finds $b \approx 30$ and $c \approx -30400$ with $\tau = \bar{\tau} = 0.3$ - a solution that satisfies all the FOC equalities $\partial W/\partial b = 0$ and $\partial W/\partial c = 0$, but is economically meaningless. Switching to NLP would not resolve this, and would have the same problem.

3.5.1 One solution

We tested a minimal fix to restore non-negativity on one instrument while leaving the other free. We can still remove the upper-bound constraints $b \leq \tau$ and $c \leq 1 - \tau$.

Two natural sub-variants arise from the economics:

1. $b \geq 0$ and c free
2. b free and $c \geq 0$.

This approach works, for example, we tried both for our chosen baseline $\alpha = 0.8, \beta = 0.5, \delta = 1.2, \bar{\tau} = 0.3, \varepsilon = 5, a = 5, \phi = 0.7$.

In (1) $\tau = \bar{\tau} = 0.3$, $b = 0$ and $c = 0.49$.

In (2) $\tau = \bar{\tau} = 0.3$, $b = -0.37$ and $c = 0.602$.

4 Shifting Model

Now we document the simulation of the Shifting Model from Section 6 of the paper.

4.1 Shifting Model (TM) reduced-form firm-response block

Following the paper's appendix, imposing firm symmetry, no cross-firm spillovers $s_i = 0$, and focusing on an individual firm level $n = 1$, the value of innovation in the Shifting Model is

$$D = \Delta^I(1 - \tau) + bP + \frac{\rho b^2}{2}P.$$

Using $P = \beta\Delta^I$, this can equivalently be written as

$$D = \Delta^I \left[(1 - \tau) + \beta b + \frac{\beta \rho b^2}{2} \right].$$

The effective input prices faced by the firm are

$$p_l = \frac{1 - \tau - c - \frac{\theta c^2}{2}}{D}, \quad p_e = \frac{1}{D}.$$

The functional form of the innovation technology is

$$\sigma = 1 - \exp(-r), \quad r = \alpha \ln l + (1 - \alpha) \ln e, \quad 0 < \alpha < 1.$$

The firm's first-order conditions (FOC) are

$$\exp(-r) \frac{\alpha}{l} = p_l, \quad \exp(-r) \frac{1 - \alpha}{e} = p_e.$$

Since these conditions require $p_l, p_e > 0$, the numerical implementation

imposes the feasibility condition $D > 0$ and

$$N_1 \equiv 1 - \tau - c - \frac{\theta c^2}{2} > 0.$$

The latter condition is tighter than the simple upper-bound constraint $c \leq 1 - \tau$, and ensures that the logarithmic input-demand system is well defined.

Solving the first-order conditions gives the closed-form input demands

$$\begin{aligned} l &= \alpha^{\frac{2-\alpha}{2}} (1 - \alpha)^{-\frac{1-\alpha}{2}} p_l^{-\frac{2-\alpha}{2}} p_e^{\frac{1-\alpha}{2}}, \\ e &= \alpha^{-\frac{\alpha}{2}} (1 - \alpha)^{\frac{1+\alpha}{2}} p_l^{\frac{\alpha}{2}} p_e^{-\frac{1+\alpha}{2}}. \end{aligned}$$

4.2 Welfare function

The reduced-form welfare function is

$$\begin{aligned} W = & \underline{\pi} + \Delta\sigma - (l + e) + \xi \left[\sigma \frac{P\rho b^2}{2} + \frac{\theta c^2}{2} l \right] \\ & + (\delta - 1) \left[\tau(\underline{\pi} + \Delta\sigma - l) - \sigma P(b + \rho b^2) - (c + \theta c^2) l \right]. \end{aligned}$$

Equivalently, defining

$$M_b \equiv P(b + \rho b^2), \quad M_c \equiv c + \theta c^2,$$

and

$$K_b \equiv \frac{P\rho b^2}{2}, \quad K_c \equiv \frac{\theta c^2}{2}.$$

$$W = \underline{\pi} + \Delta\sigma + \xi K_b \sigma + \xi K_c l + (\delta - 1) \left[\tau(\underline{\pi} + \Delta\sigma) - M_b \sigma - (\tau + M_c) l \right] - (l + e).$$

4.3 Derivative blocks

The price derivatives are

$$\frac{\partial p_l}{\partial \tau} = \frac{p_l \Delta^I - 1}{D}, \quad \frac{\partial p_e}{\partial \tau} = \frac{p_e \Delta^I}{D},$$

$$\frac{\partial p_l}{\partial b} = -\frac{p_l P(1 + \rho b)}{D}, \quad \frac{\partial p_e}{\partial b} = -\frac{p_e P(1 + \rho b)}{D},$$

and

$$\frac{\partial p_l}{\partial c} = -\frac{1 + \theta c}{D}, \quad \frac{\partial p_e}{\partial c} = 0.$$

The minimal derivative set required for the welfare gradients is

$$\frac{\partial \sigma}{\partial r} = 1 - \sigma, \quad \frac{\partial r}{\partial p_l} = -\frac{\alpha}{2p_l}, \quad \frac{\partial r}{\partial p_e} = -\frac{1 - \alpha}{2p_e},$$

$$\frac{\partial l}{\partial p_l} = -\frac{2 - \alpha}{2} \frac{l}{p_l}, \quad \frac{\partial l}{\partial p_e} = \frac{1 - \alpha}{2} \frac{l}{p_e},$$

and

$$\frac{\partial e}{\partial p_l} = \frac{\alpha}{2} \frac{e}{p_l}, \quad \frac{\partial e}{\partial p_e} = -\frac{1 + \alpha}{2} \frac{e}{p_e}.$$

Define the tax-revenue response blocks

$$A_l = (\tau \Delta - M_b) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_l} - (\tau + M_c) \frac{\partial l}{\partial p_l},$$

$$A_e = (\tau \Delta - M_b) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_e} - (\tau + M_c) \frac{\partial l}{\partial p_e},$$

and the non-tax welfare response blocks

$$B_l = (\Delta + \xi K_b) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_l} + \xi K_c \frac{\partial l}{\partial p_l} - \left(\frac{\partial l}{\partial p_l} + \frac{\partial e}{\partial p_l} \right),$$

$$B_e = (\Delta + \xi K_b) \frac{\partial \sigma}{\partial r} \frac{\partial r}{\partial p_e} + \xi K_c \frac{\partial l}{\partial p_e} - \left(\frac{\partial l}{\partial p_e} + \frac{\partial e}{\partial p_e} \right).$$

Using the closed-form derivatives, these can also be written as

$$A_l = \frac{1}{2p_l} [-\alpha(\tau \Delta - M_b)(1 - \sigma) + (2 - \alpha)(\tau + M_c)l],$$

$$A_e = -\frac{1 - \alpha}{2p_e} [(\tau \Delta - M_b)(1 - \sigma) + (\tau + M_c)l],$$

$$B_l = \frac{1}{2p_l} [-\alpha(\Delta + \xi K_b)(1 - \sigma) - \xi K_c(2 - \alpha)l + (2 - \alpha)l - \alpha e],$$

$$B_e = \frac{1}{2p_e} [-(1-\alpha)(\Delta + \xi K_b)(1-\sigma) + \xi K_c(1-\alpha)l - (1-\alpha)l + (1+\alpha)e].$$

4.4 Welfare derivatives w.r.t. τ, c, b

The derivative of welfare with respect to τ is

$$\frac{\partial W}{\partial \tau} = (\delta - 1)(\pi + \Delta\sigma - l) + (\delta - 1) \left[A_l \frac{p_l \Delta^I - 1}{D} + A_e \frac{p_e \Delta^I}{D} \right] + B_l \frac{p_l \Delta^I - 1}{D} + B_e \frac{p_e \Delta^I}{D}.$$

The derivative of welfare with respect to c is

$$\frac{\partial W}{\partial c} = \xi \theta c l + (\delta - 1) \left[-l(1 + 2\theta c) - A_l \frac{1 + \theta c}{D} \right] - B_l \frac{1 + \theta c}{D}.$$

The derivative of welfare with respect to b is

$$\begin{aligned} \frac{\partial W}{\partial b} = & \xi \sigma P \rho b + (\delta - 1) \left[-\sigma P(1 + 2\rho b) - A_l \frac{p_l P(1 + \rho b)}{D} - A_e \frac{p_e P(1 + \rho b)}{D} \right] \\ & - B_l \frac{p_l P(1 + \rho b)}{D} - B_e \frac{p_e P(1 + \rho b)}{D}. \end{aligned}$$